

LEONARDO MEDINA

Geospatial Data Analyst | Forestry Engineer

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Geospatial analyst shaped by forestry engineering: spotting spatial patterns and building data-driven solutions – from ecological monitoring to operational intelligence across industries.

I combine an environmental knowledge base, scientific methodology, and data analysis techniques to develop modern GIS workflows—Python, Google Earth Engine, QGIS/ArcGIS and spatial SQL—helping organizations solve environmental challenges and optimize operations through geospatial intelligence.

Backed by years of field experience and continuous learning: field surveys across varied ecosystems, agroforestry advisory, biodiversity inventories—documented in peer-reviewed publications— while staying current with emerging GIS technologies.

SKILLS

Geospatial Analysis & Remote Sensing: QGIS • Python (GeoPandas, Rasterio, Geemap, Shapely, Earth Engine API) • ArcGIS Pro • ArcGIS Online • Google Earth Engine • Remote sensing • Multi-temporal analysis • Geoprocessing • PostGIS • DuckDB

Programming & Automation: Python • API consumption • SQL (PostgreSQL, MySQL) • Git • ETL pipelines • Linux • Javascript/HTML/CSS

Data Visualization & Web Mapping: Leaflet • Folium • Power BI • Looker • Cartographic design • Interactive dashboards

Domain Expertise: Environmental monitoring • Biodiversity assessment • Forest inventory • Conservation planning • Scientific methodology • Statistical analysis

EDUCATION

Forestry Engineer: Universidad de Los Andes (ULA), Mérida - Venezuela | Feb 2020

EXPERIENCE

Environmental Advisor | Unidad de Respuesta Operativa (URO) - Milla River Early Warning System (*Mérida, Venezuela*) | Apr 2026 - Present | Remote

- Developing the environmental, geospatial, and hydrometeorological components for the modernization of the community-based Milla River flash-flood early warning system.
- Conducting a biophysical, geomorphological, and hydrological characterization of the watershed by integrating topographic, hydrographic, soil, lithologic, land-cover, and critical-infrastructure data.
- Automating the download, cleaning, quality control, and analysis of meteorological and geospatial data with Python.
- Producing thematic maps, technical reports, and reproducible analytical outputs.

Data Analyst | Remote | Transportation Services Client | Jun 2025 – Dec 2025

- Built ETL pipelines using Python and SQL to process data from multiple sources (Excel, CSV, PostgreSQL), organizing station-level and event data for spatial analysis
- Created map-based analysis tools with Python (Pandas, GeoPandas, Folium) to explore passenger flow, station activity, revenue distribution, and service incidents across cities
- Developed Power BI dashboards combining spatial and operational data, helping identify demand patterns and performance issues by location

Data Analytics & Automation Specialist | Remote | Odontology Coach: Feb 2025 – Jan 2026

- Designed and automated 15+ KPI tracking systems across dental clinics, structuring data to support performance analysis across locations
- Built dashboards and automated reporting workflows using near real-time data on patient visits, treatments, and campaign performance
- Implemented data validation and transformation processes to improve data consistency and support reliable segmentation and trend analysis

Botanical Research Consultant | Jardín Botánico de Mérida, Universidad de Los Andes | Apr 2019 – Dec 2022

- Conducted floristic surveys generating spatial datasets for 200+ tree and shrub species

- Consolidated ecological, geographical, and ethnobotanical data from scientific literature for these species, integrated it into **3 peer-reviewed publications**: two plant catalogs ([trees](#) and [shrubs](#)) and a tree [identification key](#)
- Developed geospatial databases and thematic maps in QGIS visualizing species distribution patterns across thematic garden zones

Environmental Data Consultant | Fundación Programa Andes Tropicales | Oct - Dec 2021

- Conducted field surveys across high-altitude Andean watersheds, collecting and structuring data for 60+ vascular plant species
- Analyzed floristic distribution patterns across elevation gradients and in relation to proximity to water bodies.
- Developed a georeferenced dataset, botanical reports, and thematic maps in ArcGIS to support conservation planning.

Agroforestry Research Assistant | **Plant Ecophysiology Lab** | **Universidad de Los Andes** | 2015 - 2019

- Conducted geospatial analysis for agricultural plot design, land use planning and irrigation systems design
- Provided technical assistance to agricultural producers on nursery design, plantation layout, fertilization planning, and crop management
- Collected and analyzed experimental data from agroforestry (timber species + cacao) and greenhouse (lettuce, strawberry, bell pepper) plant physiology research projects

CERTIFICATIONS

- [Google Data Analyst](#) | Google. 2025
- [Visualizing Land Cover and Land Use Change with NASA Satellite Imagery](#) | NASA ARSET. 2026
- [Data Analysis with Python](#) | IBM. 2025
- [Power BI](#) | Santander Bank - Open Academy. 2025
- [Interactive SQL](#) | Desafío Latam Academy. 2025
- [Planning for Renewable Natural Resources and the Environment](#) | (CIDIAT-ULA). 2024

PUBLICATIONS

- Soto, C., Medina Linares, J. L., & Gámez, L. E. (2024). [Catálogo dendrológico del Jardín Botánico de Mérida, estado Mérida, Venezuela](#). Universidad de Los Andes. ISBN 978-980-11-2149-7.
- Soto, C., & Medina Linares, J. L. (2023). [Plantas de utilidad común del Jardín Botánico de Mérida](#). Revista de la Facultad de Farmacia. DOI: 10.53766/REFA/2024.65.2.03.
- Medina Linares, J. L., Soto, C. A., & Gámez, L. E. (2021). [Clave de identificación de árboles del Jardín Botánico de Mérida por medio de caracteres vegetativos](#). Pittieria, 44, 104-127.

LANGUAGES

Spanish: Native | **English:** C1 - Professional working proficiency